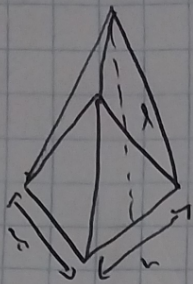
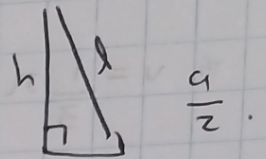


square base pyramid



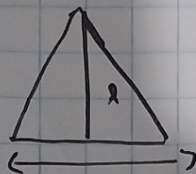
no of square = 9.

$$\text{square} = a \times a = a^2.$$



$$l^2 = h^2 + \left[\frac{a}{2}\right]^2.$$

$$\text{volume} = \frac{1}{3} a^2 h.$$



$$\text{Area} = \frac{1}{2} \times a \times l = \frac{a l}{2}$$

$$45 = \frac{a l \times 4}{2} \\ = 2 a l.$$

$$\text{total surface} = a^2 + 2 a l.$$

⑦

a) the radius of the base of a solid cylinder is $2a$ and its height is $3h$.

I) Find the area of the curved surface of the cylinder in terms of 2π , a and h .

II. Indicate the volume of that cylinder in terms of π , a and h .

$$\text{I. CSA} = 2\pi r h.$$

$$= 2\pi (2a) (3h) = 2 \times \pi \times 2a \times 3h.$$

$$= 12\pi a h.$$

$$= 12\pi a h.$$

$$\text{II. } V = \pi r^2 h.$$

$$\text{II. } V = \pi r^2 h.$$

$$= \pi (2a)^2 \times 3h.$$

$$4 \times 3 = 12$$

$$= 12\pi a^2 h.$$

b) The radius of the base of a right circular cone is $2a$ and its height is $2a$ find its volume in terms of π and a .

$$V = \frac{1}{3} \pi r^2 h$$

$$r^2 = 4a^2$$

$$\text{volume} = \frac{1}{3} \times \pi \times 4a^2 \times 2a$$

$$4 \times 2 = 8$$

$$\text{volume} = \frac{8}{3} \pi a^3$$

c) the radius of the base of a right circular cone is $2a$ and its height is $2a$. find its volume.

I.

If the cylinder and the cone are made of the same materials, indicate the volume of the materials from which the cylinder and the cone are made in terms of π and a when $h = 2a$.

II. The cylinder and the cone are melted and spheres of radius a are made without wastage. find the number of spheres that are made.

$$I. 12\pi a^2 h$$

$$12\pi a^2 \times 2a$$

$$12 \times 2 = 24 = 24\pi a^3$$

$$\text{cone } V = \frac{8}{3} \pi a^3$$

$$\text{total } V = 24\pi a^3 + \frac{8}{3} \pi a^3$$

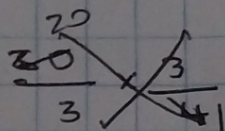
$$24 = \frac{72}{3}$$

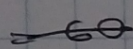
$$\frac{72}{3} \pi a^3 + \frac{8}{3} \pi a^3$$

11.

~~volume~~ of one sphere = $\frac{4}{3} \pi a^3$

$$\frac{80}{3} \pi a^3 \div \frac{4}{3} \pi a^3$$





$$\frac{80}{3} \times \frac{3}{4} = \frac{80}{4} = 20 //$$

2